Simultaneous Group-Envelope Bounds for Γ-Robust Multiple-Choice Knapsack Problems

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July 2026

Abstract

Budgeted uncertainty reduces a binary optimization problem to nominal subproblems indexed by deviation thresholds, but processing every threshold relaxation can dominate computation. For a robust multiple-choice knapsack problem, exactly-one groups permit simultaneous evaluation. Dualizing a fixed-threshold capacity constraint cancels threshold-dependent baselines, leaving the maximum of a constant and a common-slope line within each local deviation interval. After sorting, prefix and suffix maxima with range accumulation evaluate one multiplier at every threshold in time linear in the threshold count and nearly linear in the option count. The resulting minimax interval bound is valid by weak duality, exact on feasible singleton intervals, and no weaker than the bounded-threshold group-clique relaxation. A convex bracketing algorithm gives the deployed bound an explicit minimization certificate, so it inherits this dominance up to a prescribed tolerance. In a fixed-design 60-instance comparison, both methods reach tolerance; the certified envelope method wins every timing and achieves a 2.33-fold geometric-mean speedup. Validation against exact epigraph LPs and a nine-instance panel transferred from a published robust-knapsack archive support correctness and out-of-generator performance.

Keywords: robust optimization; multiple-choice knapsack; parametric Lagrangian bounds
Area of review: Optimization.
Subject classifications: Programming, integer: robust multiple-choice knapsack; Programming, linear: Lagrangian upper bounds; Analysis of algorithms: parametric evaluation.

1 Introduction

Cardinality-budget uncertainty reduces a robust binary problem to finitely many nominal problems indexed by a scalar protection threshold (Bertsimas and Sim, 2003, 2004). The reduction is polynomial, but a conventional method may still solve one nominal relaxation for every distinct deviation. Strong bounded-threshold formulations and divide-and-conquer reduce this burden, yet interval relaxation solves can dominate on large instances (Álvarez-Miranda et al., 2013; Hansknecht et al., 2018; Büsing et al., 2023).

Many operational decisions have an exactly-one menu structure: one price per product, one configuration per component, or one service level per customer segment. Every fixed-threshold relaxation of the resulting robust model is a multiple-choice knapsack linear program. Binary and at-most-one knapsacks are included by adding a dummy option. Although one fixed relaxation is classical and inexpensive, jointly bounding the full parameterized family is a different computational task.

This paper identifies a cancellation created by the exactly-one equations. After dualizing a fixed-threshold capacity constraint, the threshold-dependent group baselines disappear. Within each local deviation interval, a group contributes the larger of a constant and a line whose slope is common to all active options. Prefix and suffix maxima together with range accumulation therefore evaluate one multiplier simultaneously over the full global threshold grid. The resulting algorithm is linear in the threshold count and nearly linear in the option count after sorting.

The paper makes four contributions. First, it derives the cancellation and a two-envelope algorithm that evaluates a multiplier over all thresholds at once. Second, it proves exact-minimax dominance over the closest bounded-threshold group-clique relaxation and supplies a convex bracketing procedure that transfers the theorem to the implemented bound up to an explicit tolerance. Third, it embeds the certified bound in an adaptive decomposition with global-bound invariants and a separately scoped exact-integer extension. Fourth, a fixed-design study separates algebra, multiplier certification, kernel scaling, matched-interval behavior, end-to-end certificate time, robustness, and external-coefficient transfer. The contribution is the simultaneous parametric bound, not a new uncertainty set, a faster algorithm for one knapsack relaxation, or universal superiority of an integer solver.

2 Robust MCKP and prior-art boundary

2.1 The Γ-robust MCKP

Let $i \in \{1, \ldots, n\}$ index groups and let J_i be the finite option set of group i. Option (i, j) has value $v_{ij} \in \mathbb{R}$, nominal resource contribution $a_{ij} \in \mathbb{R}$, and nonnegative deviation $d_{ij} \geq 0$. Exactly one option is selected from every group. With an integer uncertainty budget $\Gamma \in \{0, \ldots, n\}$, the problem is

$$
\begin{aligned}
\max_{x} \quad & \sum_{i} \sum_{j \in J_i} v_{ij} x_{ij} && \text{(R-MCKP)} \\
\text{s.t.} \quad & \sum_{i} \sum_{j \in J_i} a_{ij} x_{ij} - \max_{\substack{0 \leq u_i \leq 1 \\ \sum_i u_i \leq \Gamma}} \sum_{i} u_i \sum_{j \in J_i} d_{ij} x_{ij} \geq 0, \\
& \sum_{j \in J_i} x_{ij} = 1 && (i = 1, \ldots, n), \\
& x_{ij} \in \{0, 1\}.
\end{aligned}
$$

For integer Γ, the inner maximum is the sum of the Γ largest selected deviations. The model covers a robust resource or residual constraint; the sign convention lets larger a_{ij} represent more nominal slack.

For a fixed binary selection, write $d_i := \sum_j d_{ij} x_{ij}$ for its selected deviation in group i. LP duality for the inner budget problem gives

$$\max_{0\le u\le 1,\ \mathbf{1}^\top u\le\Gamma}\sum_i d_i u_i = \min_{\theta\ge 0}\left\{\Gamma\theta + \sum_i [d_i-\theta]_+\right\}. \tag{1}$$

Define the global candidate set

$$\mathcal{B} := \{0\}\cup\{d_{ij} : i=1,\dots,n,\ j\in J_i\} = \{b_0<\cdots<b_{B-1}\}, \qquad B := |\mathcal{B}|.$$

Because the right-hand side of (1) is convex piecewise linear with breakpoints at the selected deviations, a binary selection is robust feasible if and only if there is a $\theta\in\mathcal{B}$ such that

$$\sum_i\sum_{j\in J_i} r_{ij}(\theta)x_{ij}\ge\Gamma\theta, \qquad r_{ij}(\theta) := a_{ij}-[d_{ij}-\theta]_+. \tag{2}$$

Thus (R-MCKP) is a finite union of fixed-threshold MCKPs. This classical representation is the starting point.

The exactly-one form also covers binary and at-most-one selection. Append a dummy option with $(v,a,d)=(0,0,0)$ to an at-most-one group. More specifically, the robust binary knapsack

$$\max\left\{\sum_i p_i y_i : \sum_i w_i y_i + \max_{0\le u\le 1,\ \mathbf{1}^\top u\le\Gamma}\sum_i \delta_i y_i u_i \le W,\ y\in\{0,1\}^n\right\}$$

is (R-MCKP) with two options per item: the unselected option $(0, W/n, 0)$ and the selected option $(p_i, W/n - w_i, \delta_i)$. Hence the results below include robust binary knapsack without changing asymptotic complexity.

For completeness, threshold sufficiency follows directly from convexity. For a fixed selected deviation vector q, the protection function $h(\theta)=\Gamma\theta+\sum_i[q_i-\theta]_+$ is convex and piecewise linear. Away from a selected deviation its slope is $\Gamma-|\{i : q_i>\theta\}|$, so a minimum is attained at zero or at a selected deviation, with a flat interval allowed. The global set $\mathcal{B}$ contains all such candidates.

2.2 Finite-menu pricing specialization

Finite-menu pricing supplies one operational specialization. Product i has reference price $\bar{p}_i>0$, exposure $w_i\ge 0$, and admissible prices $p_{ij}>0$, $j\in J_i$; admissibility can encode the reference band $p_{ij}\in[(1-\sigma_i)\bar{p}_i,(1+\sigma_i)\bar{p}_i]$ for $\sigma_i\in[0,1)$. Binary x_{ij} selects exactly one price. At option j, demand is $\widetilde{g}_{ij}=\hat{g}_{ij}+\xi_i\delta_{ij}$, where $0\le\delta_{ij}\le\hat{g}_{ij}$, $|\xi_i|\le 1$, and $\sum_i|\xi_i|\le\Gamma$. Let $\widetilde{N}(x,\xi) := \sum_{i,j} w_i p_{ij}\widetilde{g}_{ij}x_{ij}$ and $\widetilde{D}(x,\xi) := \sum_{i,j} w_i\bar{p}_i\widetilde{g}_{ij}x_{ij}$. Nominal revenue $\widetilde{N}(x,0)$ is maximized subject to $\widetilde{N}(x,\xi)/\widetilde{D}(x,\xi)\ge\Delta$ for every admissible ξ.

If $\widetilde{D}>0$ throughout, cross-multiplication and adverse signs subtract the Γ largest absolute selected residual deviations. The coefficients of (R-MCKP) are

$$v_{ij}=w_i p_{ij}\hat{g}_{ij}, \qquad a_{ij}=w_i(p_{ij}-\Delta\bar{p}_i)\hat{g}_{ij}, \qquad d_{ij}=\left|w_i(p_{ij}-\Delta\bar{p}_i)\delta_{ij}\right|. \tag{3}$$

Without positivity, only the additive residual interpretation remains. This specialization motivates the application-derived panel but is not required by the algorithm.

2.3 Prior-art boundary

Table 1 separates the task studied here from its ingredients: threshold enumeration and filtering (Bertsimas and Sim, 2003; Álvarez-Miranda et al., 2013; Lee and Kwon, 2014), divide-and-conquer (Hansknecht et al., 2018), robust formulations and inequalities (Atamtürk, 2006; Fischetti and Monaci, 2012; Monaci et al., 2013; Joung and Park, 2021; Joung et al., 2023), and fixed-MCKP algorithms (Sinha and Zoltners, 1979; Zemel, 1980; Dyer, 1984; Szkaliczki, 2025). The closest comparator is the bounded-threshold framework with clique aggregation of Büsing et al. (2023). None of these sources, to our knowledge, derives the baseline cancellation, two-envelope representation, certified simultaneous multiplier evaluation, or its exact-minimax dominance relation for this threshold family.

The argument proceeds in three logical steps. The threshold representation first converts the robust model into a finite family of fixed-threshold linear relaxations. Lagrangian duality then replaces that family by a minimax interval certificate and permits comparison with the group-clique relaxation. Finally, the exactly-one equations expose the cancellation that makes simultaneous evaluation computationally useful. Keeping these steps separate is important: certificate validity follows from duality, whereas speed follows from the group-envelope data structure.

3 A Lagrangian bound over threshold intervals

Fix $\theta \in \mathcal{B}$ and define the group baseline

$$r_i^*(\theta) := \max_{j \in J_i} r_{ij}(\theta), \quad c_{ij}(\theta) := r_i^*(\theta) - r_{ij}(\theta) \geq 0, \quad C(\theta) := \sum_i r_i^*(\theta) - \Gamma\theta.$$

Using the exactly-one equations, (2) is equivalent to $\sum_{i,j} c_{ij}(\theta)x_{ij} \leq C(\theta)$. If $C(\theta) < 0$, the fixed-threshold problem is infeasible. If $C(\theta) \geq 0$, choosing a maximizer of $r_{ij}(\theta)$ in every group gives zero total transformed cost, so the disjunct is feasible. Its LP relaxation value is

$$L(\theta) := \max \left\{ \sum_{i,j} v_{ij}x_{ij} : \sum_{i,j} c_{ij}(\theta)x_{ij} \leq C(\theta), \quad \sum_j x_{ij} = 1,\ x \geq 0 \right\}. \tag{4}$$

Define the feasible-threshold index set $\mathcal{F} := \{k \in \{0, \dots, B-1\} : C(b_k) \geq 0\}$ and assume $\mathcal{F} \neq \varnothing$; if it is empty, every threshold disjunct is infeasible. The target certificate is

$$M := \max_{k \in \mathcal{F}} L(b_k), \tag{5}$$

an upper bound on the integer robust optimum obtained by relaxing every disjunct. More explicitly, if Z_k^{IP} is the integer optimum of threshold disjunct k, then the classical union representation gives $Z^{\mathrm{robust}} = \max_{k \in \mathcal{F}} Z_k^{\mathrm{IP}} \leq \max_{k \in \mathcal{F}} L(b_k) = M$.

Dualize the capacity constraint of (4) with $\lambda \geq 0$:

$$D(\lambda, \theta) := \lambda C(\theta) + \sum_i \max_{j \in J_i} \{v_{ij} - \lambda c_{ij}(\theta)\}. \tag{6}$$

For a contiguous index interval $I = \{\ell, \ell + 1, \ldots, u\} \subseteq \{0, \ldots, B - 1\}$, set

$$U(I) := \inf_{\lambda \geq 0} \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k), \tag{7}$$

where $\max \varnothing := -\infty$. If interval I has a nonempty finite evaluated multiplier set $\Lambda_I \subseteq [0, \infty)$, define $\widehat{U}_{\Lambda_I}(I) := \min_{\lambda \in \Lambda_I} \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k)$.

Proposition 1 (Validity, finite-search safety, and singleton exactness)**.** *For every interval I,*

$$\max_{k \in I \cap \mathcal{F}} L(b_k) \leq U(I) \leq \widehat{U}_{\Lambda_I}(I).$$

For a feasible singleton $I = \{k\}$, $U(\{k\}) = L(b_k)$.

Proof. If $I \cap \mathcal{F} = \varnothing$, then every displayed quantity equals $-\infty$ by convention, so the claim is immediate. Assume henceforth that $I \cap \mathcal{F} \neq \varnothing$. For every $k \in \mathcal{F}$ and $\lambda \geq 0$, weak duality gives $L(b_k) \leq D(\lambda, b_k)$. Maximizing over $k \in I \cap \mathcal{F}$ and then taking the infimum over λ gives the first inequality. Restricting an infimum to Λ_I can only increase its value, giving the second. On a feasible singleton, (6) is the Lagrangian dual obtained after optimizing over the product of group simplices. The feasible bounded LP (4) satisfies strong duality, hence $L(b_k) = \inf_{\lambda \geq 0} D(\lambda, b_k)$. □

To compare the minimax bound formally with the closest prior-art relaxation, let $\underline{\theta} = b_\ell$, $\overline{\theta} = b_u$, and define $Q(I)$ as the optimum of the bounded-threshold group-clique LP

$$\begin{aligned}
\max_{x,p,z} \quad & \sum_{i,j} v_{ij} x_{ij} && \qquad (8)\\
\text{s.t.} \quad & \sum_{i,j} \Big[-a_{ij} + [d_{ij} - \overline{\theta}]_+\Big] x_{ij} + \sum_i p_i + \Gamma z \leq -\Gamma \underline{\theta}, \\
& \sum_j [\min\{d_{ij}, \overline{\theta}\} - \underline{\theta}]_+ x_{ij} \leq p_i + z && (i = 1, \ldots, n), \\
& \sum_j x_{ij} = 1 && (i = 1, \ldots, n), \\
& x, p \geq 0, \qquad 0 \leq z \leq \overline{\theta} - \underline{\theta}.
\end{aligned}$$

This is the exactly-one specialization of the bounded-threshold clique construction of Büsing et al. (2023). We use the extended-value convention $Q(I) = -\infty$ if this maximization problem is infeasible.

Theorem 2 (Dominance by the exact minimax envelope)**.** *For every threshold interval I, the exact minimax* *group-envelope bound is no larger than the bounded-threshold group-clique bound:*

$$\max_{k \in I \cap \mathcal{F}} L(b_k) \leq U(I) \leq Q(I).$$

Thus $U(I)$ is at least as strong as the comparator relaxation for the objective under study, while avoiding an interval LP solve.

Proof. If $I \cap \mathcal{F} = \emptyset$, then $U(I) = -\infty$ and both inequalities are immediate. Hence assume $I \cap \mathcal{F} \neq \emptyset$. The epigraph formulation of (7) is a linear program. Its dual has variables $\mu_k \geq 0$ and $y_{kij} \geq 0$ and can be written

$$\begin{aligned}
\max \quad & \sum_{k \in I \cap \mathcal{F}} \sum_{i,j} v_{ij} y_{kij} \\
\text{s.t.} \quad & \sum_{k \in I \cap \mathcal{F}} \mu_k = 1, \qquad \sum_{j} y_{kij} = \mu_k \qquad\qquad (k, i), \\
& \sum_{k,i,j} c_{ij}(b_k) y_{kij} \leq \sum_{k} \mu_k C(b_k).
\end{aligned}$$

Strong duality applies because the primal epigraph is feasible and has a finite optimum. The dual can be read as a convex mixture of threshold-indexed group-simplex points coupled by one aggregate capacity inequality. Take any feasible dual solution and set

$$\begin{aligned}
x_{ij} &:= \sum_k y_{kij}, & z &:= \sum_k \mu_k (b_k - \underline{\theta}), \\
p_i &:= \sum_{k,j} [\min\{d_{ij}, \overline{\theta}\} - b_k]_+ y_{kij}.
\end{aligned}$$

The simplex equations and bounds on z follow immediately. For every $b_k \in [\underline{\theta}, \overline{\theta}]$,

$$[\min\{d_{ij}, \overline{\theta}\} - \underline{\theta}]_+ \leq (b_k - \underline{\theta}) + [\min\{d_{ij}, \overline{\theta}\} - b_k]_+,$$

so the group-clique rows hold after multiplication by y_{kij} and summation. Expanding the aggregate capacity row of the dual and using $\sum_j y_{kij} = \mu_k$ cancels the group baselines and gives

$$\sum_{k,i,j} r_{ij}(b_k) y_{kij} \geq \Gamma \sum_k \mu_k b_k.$$

The identity $[d - b_k]_+ = [d - \overline{\theta}]_+ + [\min\{d, \overline{\theta}\} - b_k]_+$ then gives the first constraint of (8). The mapped point has objective $\sum_{k,i,j} v_{ij} y_{kij}$, so every feasible dual value is attained by a feasible clique-LP point. Maximizing and applying strong duality proves $U(I) \leq Q(I)$. □

The finite evaluated bound $\widehat{U}_{\Lambda_I}(I)$ remains valid by Proposition 1, but an arbitrary coarse multiplier set need not inherit Theorem 2. Theorem 6 below closes this gap by quantifying the distance between an explicitly evaluated bound and $U(I)$.

The order of maximum and infimum matters. Generally $\max_k \inf_\lambda D(\lambda, b_k) \le \inf_\lambda \max_k D(\lambda, b_k)$, so a nonsingleton interval bound can exceed the complete fixed-threshold scan. Successive splitting tightens this relaxation, and complete refinement to feasible singletons eliminates it; the evaluation algorithm below removes its computational bottleneck.

4 Simultaneous group-envelope evaluation

Theorem 3 (Baseline cancellation)**.** *For every $\lambda \ge 0$ and $\theta \in \mathcal{B}$,*

$$D(\lambda, \theta) = \sum_i \max_{j \in J_i} \{v_{ij} + \lambda r_{ij}(\theta)\} - \lambda \Gamma \theta. \tag{9}$$

Proof. Substitute $C(\theta) = \sum_i r_i^*(\theta) - \Gamma\theta$ and $c_{ij}(\theta) = r_i^*(\theta) - r_{ij}(\theta)$ into (6). For each group,

$$\lambda r_i^*(\theta) + \max_j \{v_{ij} - \lambda[r_i^*(\theta) - r_{ij}(\theta)]\} = \max_j \{v_{ij} + \lambda r_{ij}(\theta)\}.$$

Summing and retaining $-\lambda\Gamma\theta$ proves the identity. □

The cancellation is consequential because the remaining threshold dependence has only two forms. For a group i, define

$$A_i(\lambda, \theta) := \max_{j : d_{ij} \le \theta} \{v_{ij} + \lambda a_{ij}\}, \tag{10}$$

$$Q_i(\lambda, \theta) := \max_{j : d_{ij} > \theta} \{v_{ij} + \lambda (a_{ij} - d_{ij})\}, \tag{11}$$

where the maximum of an empty set is $-\infty$.

Lemma 4 (Two-envelope representation)**.** *For $\lambda > 0$,*

$$\max_j \{v_{ij} + \lambda r_{ij}(\theta)\} = \max\{A_i(\lambda, \theta), Q_i(\lambda, \theta) + \lambda\theta\}. \tag{12}$$

Between consecutive distinct deviations of group i, A_i and Q_i are constant. Hence the group contribution is the maximum of one constant and one line of slope λ on that local interval, with at most one crossover.

Proof. If $d_{ij} \le \theta$, then $r_{ij}(\theta) = a_{ij}$ and the option score is constant in θ. If $d_{ij} > \theta$, then $r_{ij}(\theta) = a_{ij} - d_{ij} + \theta$, and the score has intercept $v_{ij} + \lambda(a_{ij} - d_{ij})$ and slope λ. Maximizing separately over the two sets yields (12). Those sets are unchanged between consecutive local deviations. □

4.1 Prefix–suffix range algorithm

Sort each group by d_{ij}. Before multiplier queries, applying the same unmasked accumulation with $v_{ij} = 0$ and $\lambda = 1$ computes every $C(b_k)$ and hence $\mathcal{F}$. The global threshold indices split into maximal contiguous blocks on which the local saturated set $\{j : d_{ij} \le b_k\}$ is unchanged; all options tied at a block's left endpoint are saturated. At a fixed λ, prefix maxima of $v_{ij} + \lambda a_{ij}$ supply the constant values in (10), while reverse suffix maxima of $v_{ij} + \lambda(a_{ij} - d_{ij})$ supply the active-line intercepts in (11). If both envelopes are finite on a block, their crossover is $(A_i - Q_i)/\lambda$; a binary search finds the first global threshold at or above it. If one envelope is empty, the other applies on the entire block. Two difference arrays accumulate the resulting constant and linear pieces; cumulative sums recover the group total at all global thresholds. Finally subtract $\lambda\Gamma b_k$.

Algorithm 1 One simultaneous multiplier evaluation

Require: Global thresholds $b_0 < \cdots < b_{B-1}$; deviation-sorted groups; feasibility mask $\mathcal{F}$; $\lambda \geq 0$

if $\lambda = 0$ **then**
  Set every entry to $\sum_i \max_j v_{ij}$
  Mask indices outside $\mathcal{F}$ by $-\infty$; **return** the vector
end if
Initialize intercept and slope difference arrays of length $B + 1$
for each group i **do**
  Compute prefix maxima of $v_{ij} + \lambda a_{ij}$
  Compute suffix maxima of $v_{ij} + \lambda(a_{ij} - d_{ij})$
  for each local deviation interval **do**
    Read its constant A and active-line intercept Q
    if $A = -\infty$ **then**
      Range-add $Q + \lambda b_k$ on the interval
    else if $Q = -\infty$ **then**
      Range-add A on the interval
    else
      Locate the clipped first index with $b_k \geq (A - Q)/\lambda$
      Range-add A before that index and $Q + \lambda b_k$ from it onward
    end if
  end for
end for
Cumulatively sum both arrays and subtract $\lambda\Gamma b_k$
Mask indices outside $\mathcal{F}$; return the B values

Theorem 5 (Exact evaluation and complexity)**.** *Let $K := \sum_i |J_i|$. In real arithmetic, after sorting the global thresholds and each group, Algorithm 1 returns the values in (9) on $\mathcal{F}$ and $-\infty$ elsewhere for a fixed multiplier in*

$$O(B + K\log(B + 1)) \text{ time and } O(B + K) \text{ working storage.}$$

Constructing the global threshold set and sorting the groups costs $O(K\log K)$. A direct score evaluation at every option–threshold pair takes $\Theta(BK)$ arithmetic operations; fully materializing those scores also takes $\Theta(BK)$ storage.

Proof. Lemma 4 proves exactness on every local interval, including repeated deviations and endpoints under the convention $d_{ij} \leq \theta$ for the saturated set. Each group produces at most $|J_i| + 1$ local intervals. Prefix and suffix maxima cost $O(K)$. Every local interval performs constant work and one binary search in the B global thresholds, totaling $O(K\log(B + 1))$. Moreover, setting all $v_{ij} = 0$ and $\lambda = 1$ in (9) yields $\sum_i \max_j r_{ij}(b_k) - \Gamma b_k = C(b_k)$, so the same range algorithm constructs the complete feasibility mask within the stated complexity. The cumulative sums and mask application cost $O(B)$. The sorted group arrays, local-interval records, and two difference arrays occupy $O(K + B)$ storage. □

4.2 Certified minimization over the multiplier

Write $\phi_I(\lambda) := \max_{k \in I \cap \mathcal{F}} D(\lambda, b_k)$ and define

$$H_I := \sum_i \max_{j \in J_i} \max\{|a_{ij}|, |a_{ij} - d_{ij}|\} + \Gamma \max_{k \in I} |b_k|. \tag{13}$$

The implementation brackets a minimizer before contracting the bracket. Starting from zero and a positive scale, it doubles the positive endpoint while the evaluated objective strictly decreases. Once two consecutive values are nondecreasing, the interval from zero to the latter point contains a minimizer. Golden-section comparisons then contract this interval. The algorithm returns the smallest explicitly evaluated value at a multiplier inside the final bracket; it does not use an optimizer-interpolated objective.

Theorem 6 (Certified deployed minimax bound)**.** *For every interval with $I \cap \mathcal{F} \neq \emptyset$, ϕ_I is finite, convex, piecewise linear, and H_I-Lipschitz on $[0, \infty)$. The geometric expansion above terminates after finitely many evaluations and produces a bracket $[\ell, u]$ containing a minimizer. For every evaluated $\bar{\lambda} \in [\ell, u]$,*

$$U(I) \leq \bar{U}(I) := \phi_I(\bar{\lambda}) \leq U(I) + H_I(u - \ell) \leq Q(I) + H_I(u - \ell). \tag{14}$$

Consequently, stopping when $H_I(u - \ell) \leq \eta_I$ yields a valid evaluated interval bound whose excess over the exact minimax value, and over the clique-dominance benchmark, is at most η_I. For a feasible singleton, $L(b_k) \leq \bar{U}(\{k\}) \leq L(b_k) + \eta_{\{k\}}$.

Proof. For fixed k, (9) is a sum of maxima of affine functions of λ, plus an affine term. It is therefore finite, convex, and piecewise linear. A slope of any active affine piece is $\sum_i r_{ij_i}(b_k) - \Gamma b_k$. Because $r_{ij}(b_k)$ lies between $a_{ij} - d_{ij}$ and a_{ij}, its absolute value is bounded by H_I. A finite pointwise maximum preserves convexity, piecewise linearity, and this common Lipschitz bound, proving the first claim.

In the baseline form (6), $c_{ij}(b_k) \geq 0$ and every group contains an option with zero transformed cost. Thus each $D(\cdot, b_k)$ has eventual slope $C(b_k) \geq 0$ for $k \in \mathcal{F}$. The finite maximum ϕ_I is consequently nondecreasing after its last breakpoint. Geometric expansion therefore reaches consecutive endpoints $a < b$ with $\phi_I(b) \geq \phi_I(a)$. Convexity then implies that some minimizer lies in $[0, b]$; otherwise all subgradients through b would be negative, contradicting the nonnegative secant slope. Standard golden-section comparisons preserve at least one minimizer in the retained bracket.

Finally, if λ^* is a minimizer in $[\ell, u]$, Lipschitz continuity gives $0 \leq \phi_I(\bar{\lambda}) - \phi_I(\lambda^*) \leq H_I|\bar{\lambda} - \lambda^*| \leq H_I(u - \ell)$. Substitute $\phi_I(\lambda^*) = U(I)$ and apply Theorem 2; singleton exactness follows from Proposition 1. □

After bracketing, golden-section contraction needs $O(\log(H_I(u - \ell)/\eta_I))$ simultaneous evaluations. The released solver uses target $\eta_I \leq 10^{-9} + 10^{-8} \max\{1, |\bar{U}(I)|\}$. Its padded rectangular group arrays require $O(B + nm_{\max} + K)$ storage, reducing to $O(B + K)$ in the equal-menu experiments. The returned evaluated value is advanced by one binary64 representable number toward $+\infty$, while the corresponding lower certificate is rounded once toward $-\infty$; the reported gap includes both steps. The theorem concerns real arithmetic; binary64 feasibility and termination use stated tolerances, so the implementation is a solver-tolerance certificate rather than an interval-arithmetic proof.

5 Adaptive certification

Let E be the set of thresholds whose fixed LPs have been solved, initialized so that $E \cap \mathcal{F} \neq \emptyset$; the feasibility mask supplies such an index whenever $\mathcal{F} \neq \emptyset$. Let $\mathcal{P}$ be a collection of active intervals covering every index in $\mathcal{F} \setminus E$; its intervals may also contain evaluated indices. Let U^{disc} be the maximum last valid bound of every removed interval not yet dominated by the incumbent, with $U^{\mathrm{disc}} = -\infty$ initially. Maintain

$$\mathrm{LB} := \max_{k \in E \cap \mathcal{F}} L(b_k), \qquad \mathrm{UB} := \max \left\{ \mathrm{LB}, U^{\mathrm{disc}}, \max_{I \in \mathcal{P}} \bar{U}(I) \right\},$$

where a maximum over an empty collection is $-\infty$. The algorithm begins with the root interval, evaluates its endpoints and midpoint, and repeatedly bisects the active interval with largest bound. The same oracle-independent endpoint–midpoint choices are evaluated in every new nonsingleton interval for both compared methods. If the three root candidates contain no feasible threshold, the initialization requirement above calls for one additional index from the already constructed feasibility mask before any gap test. All timed comparison instances had a feasible endpoint or midpoint, so this fallback was not invoked there. An interval can be removed once its stored bound is within the requested tolerance of LB; its last valid upper bound is retained in U^{disc} until LB dominates it. Singleton thresholds are solved directly.

Proposition 7 (Certificate invariant)**.** *At every iteration,* $\mathrm{LB} \leq M \leq \mathrm{UB}$. *If the procedure stops when*

$$\mathrm{UB} - \mathrm{LB} \leq \varepsilon \max\{1, |\mathrm{LB}|\},$$

then it certifies (5) to scaled absolute–relative tolerance ε. If all feasible thresholds have been evaluated, then $\mathrm{LB} = M$ *and all interval records may be removed, setting* $\mathrm{UB} := \mathrm{LB}$. *Alternatively, if every active and retained discarded bound is already at most* LB, *then the maintained bounds satisfy* $\mathrm{LB} = \mathrm{UB} = M$.

Proof. LB is the maximum over a subset of fixed-threshold LP values. Every unevaluated feasible threshold belongs to an active interval or to a removed interval whose retained bound is represented by U^{disc}; Proposition 1 therefore implies $M \leq \mathrm{UB}$. Splitting replaces a parent by children whose union is the parent, and direct singleton evaluation transfers its exact LP value into LB. These operations preserve the inequalities. The stopping claim follows by division by $\max\{1, |\mathrm{LB}|\}$. If every feasible threshold has been evaluated, the definition of M gives $M = \mathrm{LB}$ and obsolete interval records can be deleted. Under the alternative bound-dominance condition, the displayed definition of UB gives $\mathrm{UB} = \mathrm{LB}$; in either case the stated equality follows. □

The primary certificate concerns M, the complete threshold-disjunctive LP upper bound. The released implementation also provides an exact integer variant: it replaces the LP lower bound by a globally robust integer incumbent, solves a fixed-threshold MCKP branch-and-bound whenever a singleton must be resolved, and prunes a whole interval whenever its retained LP upper bound cannot improve the incumbent.

Corollary 8 (Exact integer integration)**.** *The integer interval-search variant maintains* $Z^{\mathrm{incumbent}} \leq Z^{\mathrm{robust}} \leq \mathrm{UB}$. *If it exhausts the active intervals or terminates with* $\mathrm{UB} - Z^{\mathrm{incumbent}}$ *within the prescribed scaled tolerance, the incumbent is globally optimal to that tolerance.*

Proof. Every fixed-threshold integer optimum is bounded by its fixed-threshold LP value, and Proposition 1 bounds all such LP values in an active interval. Direct singleton solution transfers an exact integer value into the incumbent; splitting preserves interval coverage; and pruning occurs only when a valid retained upper bound cannot improve the incumbent. The invariant and stopping claim follow exactly as in Proposition 7. □

6 Computational study

6.1 Design and comparators

The certified-minimization protocol was serialized before the final rerun; its SHA-256 digest is a63a593f1c67 db021624a75a241e4b44123982222c7830a6095dd9c22b8a98f2. The factorial instance design predates the certified oracle, and the complete final protocol is serialized with the release; it is a reproducible fixed design rather than an external preregistration. All runs used one thread on an Apple M4 under macOS, Python 3.14.2, NumPy 2.4.4, and SciPy 1.17.1 with HiGHS. Source data, raw timing repetitions, environments, summaries, and generation scripts accompany the paper.

Four structured families isolate distinct difficulties: *dense frontier* gives many competitive value–resource tradeoffs; *correlated risk* couples value gains and deviations; *near tie* produces unstable rankings; and *many breakpoints* gives essentially unique deviations, so B grows with K. The primary panel uses six options per group, $\Gamma = \lfloor \sqrt{n} \rfloor$, $n \in \{360, 720, 1440\}$, five seeds per family–size cell, and five timing repetitions per method. Method order alternates within an instance; the instance median is the sole statistical observation. We report geometric-mean paired speedup and a 10,000-draw design-stratified bootstrap interval that resamples seeds within each family–size cell. These descriptive summaries quantify variation across the fixed generated design; they do not define a superpopulation of robust MCKPs.

The end-to-end comparator is the bounded-threshold LP of Büsing et al. (2023), specialized with one clique/GUB inequality per exactly-one group as in (8) and assembled in sparse CSR form. Both methods use the same best-first splitting policy, fixed-threshold LP routine, endpoint–midpoint candidate rule, scaled tolerance $\varepsilon = 10^{-6}$, and time limit including preprocessing. They differ only in the interval upper bound: the comparator solves a bounded-threshold clique LP, whereas the compressed method evaluates the group-envelope Lagrangian bound.

The shared fixed-threshold routine constructs each group's upper value–cost hull and greedily merges its slopes, avoiding branch-and-bound data that neither certificate needs. Held-out validation compares this LP-only path with an independent reference routine at every threshold; both certificates therefore use the same fixed-MCKP solver.

Protocol-fixed gates required maximum algebraic error at most 2×10^{-6}; kernel speedup at least 3× for $n \geq 360$; 100% primary tolerance completion; geometric-mean primary speedup at least 2× with interval lower endpoint above 1.5×; at least 80% timing wins; root dominance in at least 95% of cases; a median win in all four families; and a median win in every robustness configuration.

6.2 Correctness validation and oracle ablations

Forty held-out irregular instances use unequal menu sizes, repeated and near-repeated deviations, signed objectives, and Γ values from 0 to n. We compare the compressed trace with (i) the dense baseline-cost trace, (ii) direct evaluation of (9), (iii) independently optimized interval bounds, (iv) exact epigraph LP solutions for randomly selected intervals, (v) an independent fixed-MCKP LP routine at every threshold, and (vi) a complete threshold scan. The maximum algebraic discrepancy is 5.59×10^{-8}; the maximum violation of the certified upper/lower enclosure against the exact epigraph LP is 0.00; and the largest scaled minimization certificate is 1.00×10^{-8}. The minimum root-bound slack over the complete scan is 0.00. All serialized validation gates pass.

The kernel ablation separates algebraic compression from adaptive search. It contains 48 instances, nine normalized multiplier queries, five repetitions, and balanced order. Table 2 reports geometric-mean query and total construction-plus-query speedups. The maximum identity error is 9.78×10^{-9}. For $n \geq 360$, total speedup is 20.73× in geometric mean. In the many-breakpoints family, where B grows linearly with K, the descriptive log–log slopes are 0.92 for the compressed evaluator and 2.02 for dense materialization; the left panel of Figure 1 shows the separation. Storage denotes persistent numerical-array bytes rather than peak process memory and grows sharply only in that family.

To isolate the interval oracles from adaptive-tree effects, a common-trace ablation evaluates both bounds on the same 168 dyadic intervals from 24 independently generated instances. The envelope oracle wins 24 instance timings with a 2.59× geometric-mean speedup; every evaluated bound is certified, with maximum scaled minimization gap 9.56×10^{-9}, and its bound is no larger than the clique bound on 100.0% of matched intervals. This comparison includes preprocessing and sparse LP assembly.

6.3 Primary result

Both methods reach tolerance on all 60 primary instances. Every envelope bound is certified, with maximum scaled minimization gap 9.43×10^{-9}. The envelope method wins all 60 paired timings, with geometric-mean speedup 2.33× and design-stratified 95% interval [2.17, 2.52]. It evaluates a median 28.8% of thresholds as fixed LPs. Final lower bounds are identical at recorded precision, and its root bound never exceeds the clique root bound within numerical tolerance. All 300/300 repeat blocks are wins (minimum 1.19×); median within-instance coefficients of variation are 1.86% and 1.65%. Across an instance's interval solves, the clique comparator uses a median 100 394 sparse-matrix nonzeros and 1.19 MiB of cumulative CSR arrays. The right panel of Figure 1 displays instance- and size-level scaling.

Every family median exceeds one, from 1.37× for many breakpoints to 2.25× for correlated risk, so the protocol-fixed breadth gate passes.

6.4 Robustness, stress, and published-coefficient evidence

The 36-instance robustness panel changes one dimension at $n = 720$: $\Gamma = 1$, twelve-option menus, or $\Gamma = \lfloor 0.2n \rfloor$. Every instance reaches tolerance, and the compressed method wins all 36 timings; every configuration-level bootstrap interval remains above one. The larger fraction of fixed thresholds under the

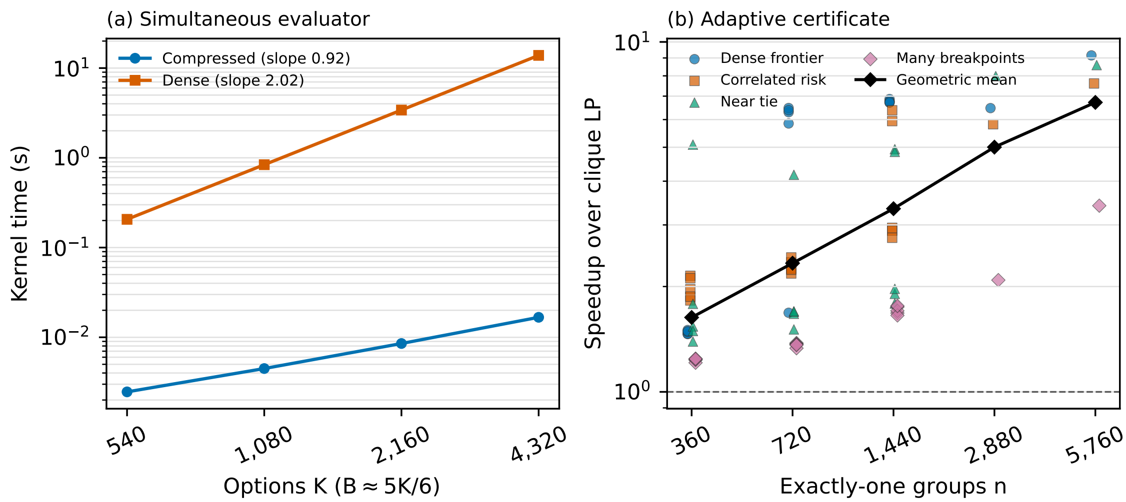

Figure 1. Mechanism and end-to-end scaling. Left: construction plus nine multiplier queries in the many-breakpoints kernel panel, where B grows linearly with K; lines show geometric means over three seeds, with descriptive log–log slopes. Right: instance-level speedup of the compressed adaptive certificate over the clique-LP certificate; black diamonds are geometric means by size. Primary observations use five seeds at $n \le 1440$ and stress observations use one separate seed at $n \in \{2880, 5760\}$.

large budget partly explains its lower speedup than the wide-menu panel.

The descriptive stress panel uses four families, one new seed, and two repetitions at each of 2,880 and 5,760 groups. All eight instances reach tolerance with identical final lower bounds. The geometric-mean speedup is 5.80× overall and 6.72× at 5,760 groups. The many-breakpoints instances contain 14,401 and 28,801 thresholds, respectively; detailed rows are in the companion.

For an out-of-generator check, we use the published robust-knapsack archive of Gersing et al. (2022). Each binary item becomes the two-option group given after (R-MCKP). We retain the archive's nominal profits, weights, budgets, and deviation vectors, but transfer its objective deviations to resource deviations; this is a transparent model-compatible coefficient stress test, not a reproduction of the source uncertain-objective model. The panel contains three published seeds at each of 1,000, 5,000, and 10,000 items. Both methods reach tolerance, every envelope bound is certified with scaled gap at most 9.85×10^{-9}, and the method wins 9/9 timings with geometric-mean speedup 2.32×, rising to 3.08× at 10,000 items.

6.5 Exact integer integration audit

The integer integration in Corollary 8 was audited separately from the LP-certificate experiment. Twelve instances use 30 groups, all four families, three new seeds, two balanced timing repetitions, and a five-second limit. The envelope and clique interval searches use the same fixed-threshold branch-and-bound and certify 11 of 12 instances each; complete threshold enumeration and compact SCIP certify all 12. Whenever at least two methods certify, their objectives differ by at most 1.14×10^{-13}. On the 11 jointly certified envelope–clique cases, the geometric-mean clique/envelope time ratio is 0.97, showing that fixed-threshold integer search, rather than interval-bound computation, dominates at this scale. The many-breakpoints family remains the decisive limitation: enumeration and compact SCIP certify all three instances while each interval method certifies two. This audit validates the exact integration and its accounting invariant, but it does not support a claim of universal integer-solver superiority; the paper's computational claim remains acceleration of the complete LP-bound certificate.

6.6 Application-derived semi-synthetic panel

The pricing model in Section 2.2 also supplies an application-derived realism check. We use the UCI Online Retail data set, which contains 541,909 transaction records (Chen, 2015). The released calibration reads the first 200,000 records; after positive-price/quantity filtering and the eight-observation SKU rule, 192,451 observations contribute to 2,549 SKU aggregates. These aggregates calibrate the segment mix, price levels, volumes, and uncertainty scales of nine semi-synthetic pricing portfolios; documented elasticity and menu-generation assumptions remain modeled rather than observed. The panel uses 360, 720, and 1,440 groups, twelve price choices, three new seeds per size, and three balanced repetitions. Both methods reach tolerance, and the compressed method wins 9/9 instances with geometric-mean speedup 1.51× (95% cell-stratified bootstrap interval [1.27, 1.85]), rising to 1.75× at 1,440 groups. This panel tests scale and coefficient realism; demand estimation is outside the algorithmic scope, and the nonrandom row cap and modeled demand curves preclude causal or transaction-level validation claims.

6.7 Scope and limitations

The experiments validate the envelope identity, certified minimax approximation, retained LP certificate, and speed advantage over the tested sparse clique-LP implementation, not universal integer-solver dominance. The public-data panel transfers coefficients rather than reproducing the source uncertain-objective problem, and the pricing panel remains semi-synthetic. The implementation uses Python/SciPy on one architecture, while compiled code could shift constants; streaming could reduce dense-oracle storage without changing its $\Theta(BK)$ arithmetic. The official implementation of Büsing et al. (2023) requires Gurobi and targets objective uncertainty, so the comparison isolates its bounded-threshold clique relaxation under the same search rather than claiming a direct reproduction of DnC+. The exact audit shows that fixed-threshold integer search can erase the interval-oracle advantage and that compact SCIP remains preferable on many-breakpoints instances. DnC+, specialized fixed-MCKP solvers, and the oracle are therefore complementary.

7 Conclusion

After sorting, exactly-one group envelopes evaluate one multiplier over all thresholds in $O(B+K\log(B+1))$ time and $O(B+K)$ working storage. Convex bracketing converts those evaluations into an explicit minimization certificate: the deployed bound is valid, within a prescribed tolerance of the exact minimax value, singleton-exact to that tolerance, and no larger than the clique bound plus that tolerance. The fixed-design study records zero certificate violations, 60 of 60 primary wins, a 2.33× geometric-mean speedup, and 9 of 9 wins on published knapsack coefficients. The exact integration also locates the boundary: integer subproblem work can dominate, and compact SCIP remains stronger on many-breakpoints instances. Multiple robust constraints remain the clearest theoretical extension.

Data and code availability

Code and reproducibility artifacts are available at https://github.com/eric939/robust_mckp. The evidence directory contains the serialized protocol, per-phase environments, raw CSV files, JSON summaries, and the SHA-256 artifact manifest.

Acknowledgments

The author thanks Angela Kunoth for her guidance and Christian Mollet and colleagues for helpful discussions.

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Table 1. Boundary between the present task and related algorithmic streams.

Stream	Established operation	Treatment here
Bertsimas–Sim and filtered enumeration	Solve a finite, possibly reduced set of nominal threshold problems	Simultaneously evaluates a Lagrangian multiplier over all thresholds
Fast fixed MCKP LP	Solves one MCKP relaxation using hull or partition structure	Amortizes evaluation across a parameterized family; does not improve the one-LP complexity
Divide-and-conquer	Avoids threshold solves using monotonicity	Replaces an interval LP by a direct exactly-one group-envelope bound
Strong and bounded-threshold formulations	Strengthen or linearize the robust model over a threshold interval	Proves objective-bound dominance over the group-clique interval LP without introducing a new robust formulation
Ellipsoidal robust MMKP (Caserta and Voß, 2019)	Treats multiple resources and covariance-based uncertainty	Different uncertainty set, geometry, and computational target

Table 2. Kernel ablation against dense threshold materialization. Speedups are geometric means over four families and three seeds. Storage is the median (maximum) ratio of persistent numerical-array bytes and excludes interpreter overhead.

Groups	Cases	Query speedup	Total speedup	Storage ratio	Maximum error
90	12	5.76	10.92	2.58 (35.3)	1.7e-10
180	12	8.69	14.60	2.60 (70.5)	5.8e-10
360	12	11.75	18.64	2.61 (140.9)	2.3e-09
720	12	14.67	23.04	2.62 (281.6)	9.8e-09

Table 3. Primary end-to-end certificate comparison. Each reported time and evaluation count is the median across instances of that size after taking the within-instance median over five balanced timing repetitions. Aggregate speedup treats each instance once. C and Q denote the compressed and clique certificates.

Groups	Cases	Compressed (s)	Clique (s)	Geometric speedup	Wins	Fixed-threshold LPs C/Q	Clique interval LPs
360	20	0.101	0.166	1.63	20/20	12/12	8
720	20	0.186	0.373	2.33	20/20	12/12	8
1440	20	0.364	0.885	3.34	20/20	12/12	8

Table 4. Robustness configurations at 720 groups, four families, and three new seeds. Within each configuration, bootstrap intervals resample the three seeds separately by family.

Configuration	Geometric speedup [95% interval]	Wins	Thresholds evaluated
Sparse budget ($m = 6, \Gamma = 1$)	1.83 [1.62, 2.28]	12/12	11.5%
Wide menu ($m = 12, \Gamma = \lfloor\sqrt{n}\rfloor$)	2.84 [2.27, 3.55]	12/12	28.8%
Large budget ($m = 6, \Gamma = 0.2n$)	2.79 [2.58, 3.12]	12/12	34.6%